2.11.1 Problems

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
–1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i =$ 5	$\sum y_i =$ 10	$\sum(x_i-\bar{x}) =$ 0	$\sum(x_i-\bar{x})^2 =$ 10	$\sum(y_i-\bar{y}) =$ 0	$\sum(x_i-\bar{x})(y_i-\bar{y}) =$ 8

$\bar{x} = 1$ $\quad \bar{y} = 2$

a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?

b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.

c. Compute $\sum_{i=1}^{5} x_i^2$, $\sum_{i=1}^{5} x_i y_i$. Using these numerical values, show that $\sum(x_i-\bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum(x_i-\bar{x})(y_i-\bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

d. Use the least squares estimates from part (b) to compute the fitted values of y, and complete the remainder of the table below. Put the sums in the last row.
Calculate the sample variance of y, $s_y^2 = \sum_{i=1}^{N}(y_i-\bar{y})^2/(N-1)$, the sample variance of x, $s_x^2 = \sum_{i=1}^{N}(x_i-\bar{x})^2/(N-1)$, the sample covariance between x and y, $s_{xy} = \sum_{i=1}^{N}(y_i-\bar{y})(x_i-\bar{x})/(N-1)$, the sample correlation between x and y, $r_{xy} = s_{xy}/(s_x s_y)$ and the coefficient of variation of x, $CV_x = 100(s_x/\bar{x})$. What is the median, 50th percentile, of x?

$1.2+0.8x_i$ $\quad y_i-\hat{y}$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
–1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i =$ 5	$\sum y_i =$ 10	$\sum \hat{y}_i =$ 10	$\sum \hat{e}_i =$ 0	$\sum \hat{e}_i^2 =$ 3.6	$\sum x_i\hat{e}_i =$ 0

e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.

f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.

g. Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.

h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i/N$.

i. Compute $\hat{\sigma}^2$.

j. Compute $\widehat{\text{var}}(b_2|\mathbf{x})$ and $\text{se}(b_2)$.

(b) $b_2 = \dfrac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\Sigma(x_i-\bar{x})^2} = \dfrac{8}{10} = 0.8$

$b_1 = \bar{y} - b_2\bar{x} = 2 - 0.8\times 1 = 1.2$

$\Rightarrow \hat{y} = 1.2 + 0.8x$

b_1: 代表 $x=0$ 時, $\hat{y}=1.2$

b_2: 代表 x 每增加一單位, $\hat{y}$ 增加 0.8

(c) x_i^2: 9, 4, 1, 1, 0

$x_i y_i$: 12, 4, 3, -1, 0

$\sum_{i=1}^{5} x_i^2 = 15$, $\sum_{i=1}^{5}(x_i-\bar{x})^2 = 15 - 5\times 1^2 = 10$
$= \Sigma x_i^2 - N\bar{x}^2$ *

$\sum_{i=1}^{5} x_i y_i = 18$, $\sum_{i=1}^{5}(x_i-\bar{x})(y_i-\bar{y})$
$= 18 - 5\times 1\times 2 = 8$
$= \Sigma x_i y_i - N\bar{x}\bar{y}$ *

(d) $(y-\bar{y})^2$: 4, 0, 1, 1, 4

$\sum_{i=1}^{5}(y-\bar{y})^2 = 10$

$\Rightarrow S_y^2 = \dfrac{\sum_{i=1}^{5}(y_i-\bar{y})^2}{5-1} = \dfrac{10}{4} = 2.5$

$S_x^2 = \dfrac{\sum_{i=1}^{5}(x_i-\bar{x})}{5-1} = \dfrac{10}{4} = 2.5$

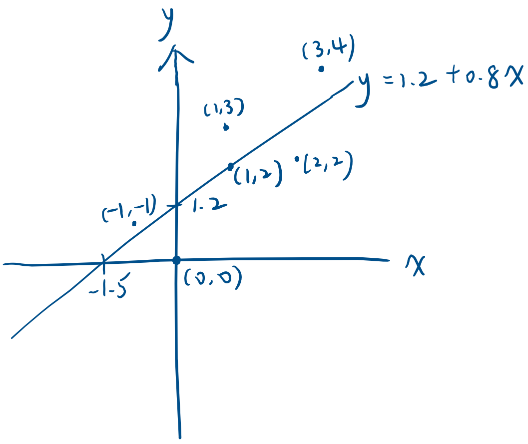
$S_{xy} = \dfrac{\sum_{i=1}^{5}(x_i-\bar{x})(y_i-\bar{y})}{5-1} = \dfrac{8}{4} = 2$

$r_{xy} = \dfrac{S_{xy}}{S_x S_y} = \dfrac{2}{\sqrt{2.5}\sqrt{2.5}} = 0.8$

$CV_x = 100 \times \dfrac{S_x}{\bar{x}} = 100 \times \dfrac{\sqrt{2.5}}{1} = 158.113883$

$x_{0.5} = 1$ *

(e)



(f) $(\bar{x}, \bar{y}) = (1,2)$

the fitted line pass through the point

(g) $\bar{y} = b_1 + b_2\bar{x} = 1.2 + 0.8 \times 1 = 2$

(h) $\bar{\hat{y}} = \dfrac{\sum_{i=1}^{5}\hat{y}}{5} = \dfrac{10}{5} = 2 = \bar{y}.$

(i) $\hat{\sigma}^2 = \dfrac{\sum_{i=1}^{5}\hat{e}_i^2}{5-2} = \dfrac{3.6}{3} = 1.2$

(j) $\widehat{Var}(b_2|x) = \dfrac{\hat{\sigma}^2}{\sum_{i=1}^{5}(x_i-\bar{x})^2} = \dfrac{1.2}{10} = 0.12$

$se(b_2) = \sqrt{\widehat{Var}(b_2|x)} = \sqrt{0.12} = 0.3464$ *

2.22 A longitudinal experiment was conducted in Tennessee beginning in 1985 and ending in 1989. A single cohort of students was followed from kindergarten through third grade. In the experiment children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star5_small*. [Note: The data file *star5* contains more observations and variables.]

a. Using children who are in either a regular-sized class or a small class, estimate the regression model explaining students' combined aptitude scores as a function of class size, $TOTALSCORE_i = \beta_1 + \beta_2 SMALL_i + e_i$. Interpret the estimates. Based on this regression result, what do you conclude about the effect of class size on learning?

b. Repeat part (a) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

c. Using children who are in either a regular-sized class or a regular-sized class with a teacher aide, estimate the regression model explaining students' combined aptitude scores as a function of the presence of a teacher aide, $TOTALSCORE = \gamma_1 + \gamma_2 AIDE + e$. Interpret the estimates. Based on this regression result, what do you conclude about the effect on learning of adding a teacher aide to the classroom?

d. Repeat part (c) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

(a)

```
Call:
lm(formula = totalscore ~ small, data = data_ab)

Residuals:
    Min      1Q  Median      3Q     Max 
-189.44  -50.44   -9.44   43.06  324.38 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  916.442      3.675 249.396   <2e-16 ***
small         12.175      5.369   2.268   0.0236 *  
---
Signif. codes:  
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared:  0.006608,	Adjusted R-squared:  0.005323 
F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

$Totalscore_i = 916.442 + 12.175\, small_i + e_i$

$\beta_1 = 916.442$ = total score in average of regular class

$\Rightarrow$ when $X=0$, $\bar{y} = 916.442$

$\beta_2 = 12.175$ = total score in average of smaller class higher than regular class

$\Rightarrow$ when X increases one unit, $\bar{y}$ increases 12.175 unit

$\Rightarrow$ The score of small class is 12.175 higher than regular class, which means there is positive effect for small class.

(b)

```
> summary(model_b_read)

Call:
lm(formula = readscore ~ small, data = data_ab)

Residuals:
    Min      1Q  Median      3Q     Max 
-74.665 -21.590  -2.665  16.410 194.335 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  432.665      1.488 290.760  < 2e-16 ***
small          6.924      2.174   3.185  0.00151 ** 
---
Signif. codes:  
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295,	Adjusted R-squared:  0.01167 
F-statistic: 10.14 on 1 and 773 DF,  p-value: 0.001507
```

```
> summary(model_b_math)

Call:
lm(formula = mathscore ~ small, data = data_ab)

Residuals:
     Min       1Q   Median       3Q      Max 
-144.777  -35.028   -5.028   29.223  142.223 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  483.777      2.449 197.545   <2e-16 ***
small          5.251      3.578   1.467    0.143    
---
Signif. codes:  
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778,	Adjusted R-squared:  0.001488 
F-statistic: 2.153 on 1 and 773 DF,  p-value: 0.1427
```

There is statistically significant evidence to suggest that small class increases reading score but not for math score, which means small class teaching may not increase score for all subjects.

(c)

```
Call:
lm(formula = totalscore ~ aide, data = data_cd)

Residuals:
     Min       1Q   Median       3Q      Max 
-191.748  -50.442   -5.442   40.558  312.558 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  916.442      3.559 257.529   <2e-16 ***
aide           4.306      4.994   0.862    0.389    
---
Signif. codes:  
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 72.23 on 835 degrees of freedom
Multiple R-squared:  0.0008898,	Adjusted R-squared:  -0.0003068 
F-statistic: 0.7436 on 1 and 835 DF,  p-value: 0.3888
```

$Totalscore_i = 916.442 + 4.306\, AIDE_i + e_i$

$\gamma_1 = 916.442$ = total score in average of regular class

$\Rightarrow$ when $X=0$, $\bar{y} = 916.442$

$\gamma_2 = 4.306$ = average total score after adding a teacher aide higher than regular class

$\Rightarrow$ when X increases one unit, $\bar{y}$ increases 4.306 unit

$\Rightarrow$ It is not significant in γ_2, that is adding a teacher aide is not significant increasing totalscore.

(d)

```
> summary(model_d_read)

Call:
lm(formula = readscore ~ aide, data = data_cd)

Residuals:
    Min      1Q  Median      3Q     Max 
-74.665 -21.536  -2.536  15.464 194.335 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  432.665      1.460 296.341   <2e-16 ***
aide           2.871      2.049   1.401    0.161    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.64 on 835 degrees of freedom
Multiple R-squared:  0.002347,	Adjusted R-squared:  0.001152 
F-statistic: 1.964 on 1 and 835 DF,  p-value: 0.1615
```

```
> summary(model_d_math)

Call:
lm(formula = mathscore ~ aide, data = data_cd)

Residuals:
     Min       1Q   Median       3Q      Max 
-146.212  -31.212   -5.777   29.223  142.223 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  483.777      2.355 205.464   <2e-16 ***
aide           1.435      3.304   0.434    0.664    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 47.79 on 835 degrees of freedom
Multiple R-squared:  0.0002258,	Adjusted R-squared:  -0.0009715 
F-statistic: 0.1886 on 1 and 835 DF,  p-value: 0.6642
```

There are not statistically significant in increaing reading score or math score after adding a teacher aide.

It is more important for improving the quality of teaching (i.e small class) than adding a teaching aide to increase students' total score.

2.25 Consumer expenditure data from 2013 are contained in the file *cex5_small*. [Note: *cex5* is a larger version with more observations and variables.] Data are on three-person households consisting of a husband and wife, plus one other member, with incomes between $1000 per month to $20,000 per month. *FOODAWAY* is past quarter's food away from home expenditure per month per person, in dollars, and *INCOME* is household monthly income during past year, in $100 units.

a. Construct a histogram of *FOODAWAY* and its summary statistics. What are the mean and median values? What are the 25th and 75th percentiles?

b. What are the mean and median values of *FOODAWAY* for households including a member with an advanced degree? With a college degree member? With no advanced or college degree member?

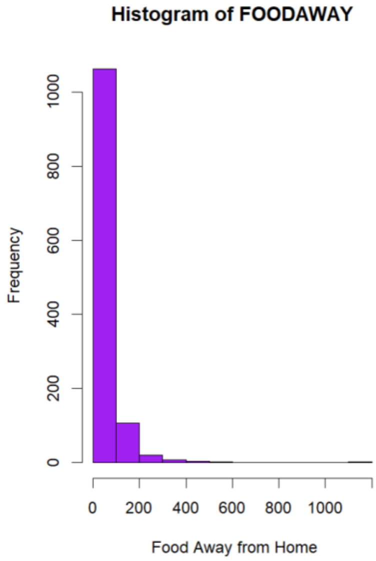
c. Construct a histogram of ln(*FOODAWAY*) and its summary statistics. Explain why *FOODAWAY* and ln(*FOODAWAY*) have different numbers of observations.

d. Estimate the linear regression $\ln(FOODAWAY) = \beta_1 + \beta_2 INCOME + e$. Interpret the estimated slope.

e. Plot ln(*FOODAWAY*) against *INCOME*, and include the fitted line from part (d).

f. Calculate the least squares residuals from the estimation in part (d). Plot them vs. *INCOME*. Do you find any unusual patterns, or do they seem completely random?

(a)




```
> summary(cex5_small$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   0.00   12.04   32.55   49.27   67.50 1179.00 
> mean(cex5_small$foodaway)
[1] 49.27085
> median(cex5_small$foodaway)
[1] 32.555
> quantile(cex5_small$foodaway, probs = c(0.25, 0.75))
    25%     75% 
12.0400 67.5025 
```

(b)

advance degree

```
> mean(cex5_small$foodaway[cex5_small$advanced == 1])
[1] 73.15494
> median(cex5_small$foodaway[cex5_small$advanced == 1])
[1] 48.15
```

college degree

```
> mean(cex5_small$foodaway[cex5_small$college == 1])
[1] 48.59718
> median(cex5_small$foodaway[cex5_small$college == 1])
[1] 36.11
```

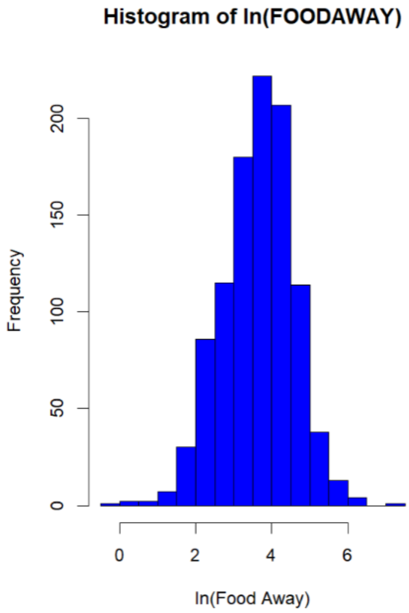
no advance or college degree

```
> mean(cex5_small$foodaway[cex5_small$college == 0 & cex5_small$advanced == 0])
[1] 39.01017
> median(cex5_small$foodaway[cex5_small$college == 0 & cex5_small$advanced == 0])
[1] 26.02
```

(c)

Observations with FOODAWAY = 0 were removed because the logarithm of zero is undefined. As a result, the number of observations for ln(FOODAWAY) is smaller than for FOODAWAY.

The histogram of ln(FOODAWAY) appears more symmetric and less skewed compared to the original FOODAWAY distribution, because taking logarithms compresses the right tail of the distribution and reduces the influence of high-spending households.




```
> summary(cex5_log$ln_foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
-0.3011  3.0759  3.6865  3.6508  4.2797  7.0724 
```

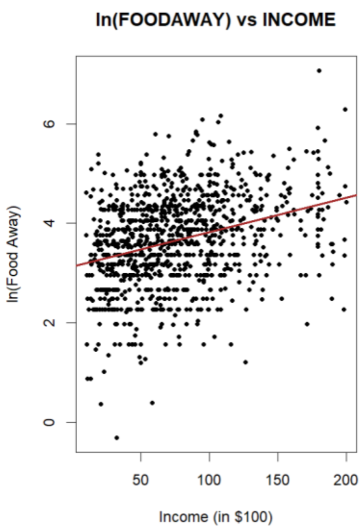
(d)

```
> summary(model_d)

Call:
lm(formula = ln_foodaway ~ income, data = cex5_log)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.6547 -0.5777  0.0530  0.5937  2.7000 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
income      0.0069017  0.0006546   10.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826,	Adjusted R-squared:  0.09738 
F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

$\widehat{\ln(\text{foodway})} = 3.1293 + 0.0069\,\text{income} + e$

⇒ As income increase one unit (i.e \$100), the expense of food way from home increase 0.69%.

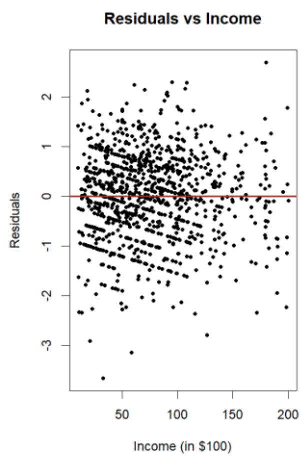
(e)



(f)

The residuals appear randomly scattered around zero, with no obvious systematic pattern.

This suggests that the linear functional form is appropriate. Although the spread of residuals seems to increase slightly as income increases, the assumption of homoscedasticity seems reasonable overall.




```
> summary(resid_d)
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
-3.65471 -0.57769  0.05302  0.00000  0.59372  2.69998 
```